

Indian Statistical Institute

End semester examination

Optimization and Numerical Methods
21 May 2025, 10:30am – 1:30pm

Bachelor of Statistical Data Science
Total points: 50

INSTRUCTIONS

- Write your name and roll number in CAPITAL LETTERS. Write your answers on the question paper itself, within the space provided.
- Answer any five questions (of 10 marks each) of six questions provided. The marks distribution for each question (among the sub-questions- (a), (b), ...) are written in the right hand side of the each questions.
- You are allowed to use your computer for computation in the examination hall, but it must be in airplane mode. Any use of the internet during the examination is strictly prohibited.
- Use computer only where the question specifically instructs you to use R/Python for computation. Report the answer in the answer sheet and there is no need to include the R/Python script in the answer sheet. You may use separate sheets for rough work, but only the completed question paper should be submitted.

Name:
Roll number:

[3+3+4]

1. (a) Draw the feasible region and find the optimal solution using graphical method for the problem,
Maximize $z = 6x + 10y$
subject to

$$3x + 5y \leq 10,$$

$$5x + 3y \leq 15,$$

$$x + y \geq 1,$$

$$x, y \geq 0.$$

- (b) An agricultural firm has 180 tons of Nitrogen fertiliser, 50 tons of Phosphate and 220 tons of Potash. It will be able to sell 3 : 3 : 4 mixtures of these substances at a profit of Rs.15 per ton and 2 : 4 : 2 mixtures of these substances at a profit of Rs.12 per ton, respectively. Formulate the linear programming problem to maximize the profit (i.e., write down the objective function and the constraints).

- (c) Using the simplex method (analytically on the paper) or using R or Python, find (and report it in the answer sheet) the quantities of the two mixtures of (b) to obtain maximum profit.

[2+2+4+2]

2. Consider developing a regression model to predict a car's fuel efficiency (MPG) based on its weight based on the following data.

Weight (1000 kg)	MPG
3.5	18
3.69	15
3.44	18
3.43	16
4.34	15
4.42	14
2.37	24

Assume a linear relationship:

$$y = \alpha + \beta \cdot x + \varepsilon$$

where y is fuel efficiency (MPG), x is weight of the car (in 1000 kg), α and β are unknown intercept and slope parameters and ε is random error following the standard normal distribution. We want to obtain the Maximum Likelihood Estimate (MLE) of α and β .

- Formulate the above as an optimisation problem to be solvable through gradient descent method for estimating the parameters α and β .
- In the answer sheet, write down the steps (or algorithm) for the above gradient descent method.

- (c) Implement the above method in R or Python and in the answer sheet report the estimates of α and β and the corresponding iteration error for the last three iterations. Take initial point: $\alpha_0 = 0$, $\beta_0 = 1$, step size: $\eta = 0.01$, maximum number of iteration=1000, threshold: $\varepsilon = 10^{-3}$.

- (d) Using Optim function in R (or in Python) compute MLE of α and β using the Nelder-Mead method. Compare the estimate of (d) with the last estimate of (c).

[3+ 2+(3+2)]

3. Consider the following data on direction of twelve animal movement as 3.91, 4.85, 2.28, 4.06, 3.70, 4.04, 5.46, 3.53, 2.28, 1.96, 2.53, 3.88. Assume that these are independent and identically distributed and are arise from the density $f(x) = [1 - \cos\{x - \theta\}]/2\pi$ on $0 \leq x \leq 2\pi$, where θ is a parameter between $-\pi$ to π . We wish to obtain MLE of θ .

(a) Derive the Newton step after formulating the above problem as an optimization problem.

- (b) Write down the algorithm (steps) for Newton's method to compute MLE for θ for step size equals to 1.

- (c) Using R or Python programming, calculate the MLE through Newton's method, with sample mean as the starting point with the error tolerance = 0.0001, maximum number of iteration as 10000. What solution do you find when you start at -2.7 and 2.7?

[3+3+3+1]

4. (a) Derive the analytical recurrence formula for the Newton-Raphson method used to approximate the root of an equation $f(x) = 0$.

- (b) Write an algorithm that computes the n^{th} root of a given positive real number N using the Newton-Raphson method. The algorithm should:
- take inputs N (a positive real number), n (a positive integer), and ε (the error tolerance),
 - returns output in each iteration showing the current approximation and the corresponding error.

- (c) Using R or Python, implement the above algorithm in part (b) to compute the **5th root of 13** with an error tolerance $\varepsilon = 0.01$. In the answer sheet, present the iteration results in tabular form.
- (d) In what situations is the Newton-Raphson method less effective than the Bisection method?

[(2+1)+4+2+1]

5. (a) Given a function f known at $(n+1)$ distinct points $x_0, x_1, \dots, x_n$, write down the Lagrange interpolating polynomial that takes the same values as f at these points. Is it essential that the x_i values must be distinct—justify?

- (b) The following is the sensor data representing the position of a moving object at specific time instances. The data is as follows:

$$\text{Time (in seconds)} = [0, 1.5, 3.0, 4.5], \quad \text{Position (in meters)} = [0, 3.4, 10.2, 18.1]$$

Derive the Lagrange interpolating polynomial that passes through these points.

(c) Use the above relation to estimate the position of the object at:

$$t = 2.0 \text{ seconds} \quad \text{and} \quad t = 4.0 \text{ seconds}.$$

(d) Can Lagrangian interpolation be used if the data points are not equally spaced? Explain.

6. (a) Derive Simpson's 1/3 Rule for approximating the definite integral

$$\int_a^b f(x) dx$$

by dividing the interval $[a, b]$ into n equal subintervals (where n is even).

- (b) Using the Trapezoidal Rule and Simpson's 1/3 Rule with $n = 4$ sub-intervals, approximate (using pen and paper) the value of

$$\int_0^4 \sqrt{x} dx.$$

- (c) Compare the approximate values obtained in part (b) with the actual integral value, and comment on the relative accuracy of both methods.

